Analytical Cume

24 October 2008

Lupus is believed to be an autoimmune disease. The cause of lupus has yet to be determined. Sometimes the immune system loses the ability to distinguish between its own body components and foreign antigens. Instead of fighting foreign antigens, the antibodies mistakenly fight the body's own cells. This is referred to as an autoimmune response and the body mistakenly makes auto-antibodies.

A local physician sends you 40-ml (milliliter) vials of blood from three different patients and asks you to test each sample for supporting evidence of a chronic disease called systemic lupus erythematosus (SLE).

You must conduct an ELISA to look for antibodies characteristic of SLE.
Note: the directions below are for patient A, but equivalent procedures are done with samples from patients B and C, and the ELISA well plate is loading accordingly.
Answer the questions following each step in the procedure.

**Step 1**
Centrifuge whole-blood samples of patients A, B, and C for 15 minutes at room temperature to get the sera

1. Why centrifuge?

**Step 2**
Using the serum from patient A, prepare three dilutions as follows:
- Take 1 ml of serum from patient A and add 1 ml of phosphate-buffered saline (PBS) solution. This is a 1:2 dilution.
- Take 1 ml of serum from patient A and add 9 ml of PBS. This is a 1:10 dilution.
- Take 0.1 ml of serum from patient A and add 9.9 ml of PBS. This is a 1:100 dilution.

2. Why make dilutions of the samples?

**Step 3**
The ELISA plate has been pretreated to bind SLE antigen to each well, place with 0.1 ml of the different dilutions of patient serum using an Eppendorf® pipettor.

3. What are the possible reactions occurring between the antigen on the plate and the sera?

**Step 4**
Add to the ELISA plate 0.1 ml dilutions for each titer of anti-DNA primary antibody (positive control) and a buffer (negative control). (Each of three wells, see diagram below)

4. Why is it necessary to add controls?

**Step 5**
Incubate the ELISA plate at 37° C for 15 minutes.

5. Why is this step necessary?

**Step 6**
Remove the fluid from each well with the pipetter and wash with 0.1 ml of PBS.

6. What is being removed during this wash? And what would happen if this step were skipped?

**Step 7**
Add 0.1 ml of buffered solution containing a secondary antibody that recognizes antibodies made in humans. Note this secondary antibody is made in a rabbit and has an attached enzyme (HRP) that will interact with the substrate in the next step. Incubate the ELISA plate at 37° C for 15 minutes.

7. Describe the chemistry going on in this step.

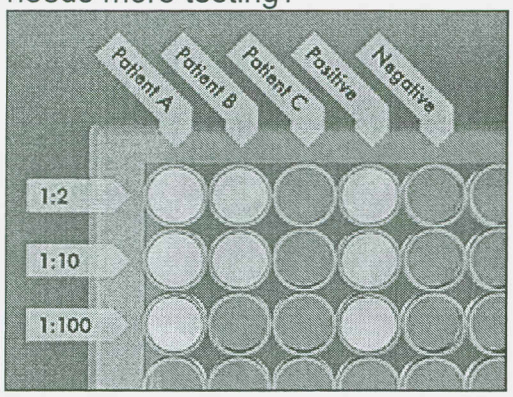
**Step 8**
Remove the fluid from each well with the pipetter and wash with 0.1 ml of PBS.

8. What is being removed during this wash? And what would happen if this step were skipped?

**Step 9**
Add 0.1 ml of buffered solution containing the chemical substrate (HRP-substrate).

9. Describe the chemistry going on here.

**Step 10**
Set timer for 15 minutes. And wait

10. Why wait? If you waited for a longer or shorter time, what would be the result?

11. From the results below, which patient is sick? which patient is not? and which needs more testing?